

CHE 435/525 *Process Systems Analysis and Control*

Lecture 12: **Frequency-Domain Dynamical Analysis**

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Recapture: Frequency response

- Frequency response: $u(t) = A \sin(\omega t) \rightarrow y(t) = \hat{A} \sin(\omega t + \phi)$
A.R. = $\hat{A}/A = |G(i\omega)|$, $\phi = \angle G(i\omega)$
- Bode diagram: A (log scale) v.s. ω (log scale), ϕ v.s. ω
- First-order system $G(s) = \frac{K}{\tau s + 1}$: a **low-pass filter and a lag**
- Lead-lag system $G(s) = \frac{1 + \tau_0 s}{1 + \tau_1 s}$: A.R. monotonic and has limits at two ends, phase angle has a max/min

Learning objectives of this lecture

- ① Calculate and explain the features of the frequency response of **non-minimum-phase** and **second-order** systems
- ② Calculate the frequency response of higher-order t.f. through the **factorization approach**
- ③ Sketch Bode diagram by **asymptotes**



Hendrik W. Bode (1905–1982)

Example 5: Non-minimum-phase (NMP) systems

- Consider t.f. (with static gain 1, denominator and numerator are conjugates)

$$G(s) = \frac{1 - \tau s}{1 + \tau s} \quad (\tau > 0) \quad \xrightarrow{s=i\omega} G(i\omega) = \frac{1 - i\omega\tau}{1 + i\omega\tau}$$

- Exercise: Show that A.R.= 1, $\phi = -2 \arctan(\omega\tau)$
- It does not “filter” any sinusoidal wave, but only causes the phase to lag

NMP system and its minimum-phase counterpart

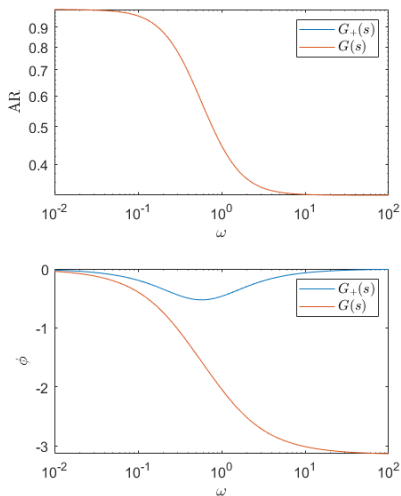
- Any system with a **RHP zero** can factorize such an element $\frac{1-\tau s}{1+\tau s} \dots$

$$G(s) = \frac{(1 - \tau s)(\dots)}{(\dots)} = \frac{1 - \tau s}{1 + \tau s} \cdot \frac{(1 + \tau s)(\dots)}{(\dots)} = \frac{1 - \tau s}{1 + \tau s} G_-(s)$$

- In $G_-(s)$, RHP zero is “reflected” from $+1/\tau$ to $-1/\tau$
- $G_-(s)$ has the same A.R. but less phase lag

$$\begin{aligned} |G(i\omega)| &= |G_-(i\omega)| \\ \angle G(i\omega) &= \angle G_-(i\omega) - 2 \arctan(\omega\tau) \end{aligned}$$

- A system with RHP zeros $G(s)$ is a **non-minimum-phase** system, and $G_-(s)$ is its **minimum-phase counterpart** without RHP zeros



$$G(s) = \frac{1-s}{1+3s}, \quad G_-(s) = \frac{1+s}{1+3s}$$

Example 6: Overdamped second-order systems

$$G(s) = \frac{K}{(1 + \tau_1 s)(1 + \tau_2 s)} \quad (K > 0, \tau_1 > \tau_2 > 0) \quad \xrightarrow{s=i\omega} G(i\omega) = \dots$$

- Evaluate modulus and angle:

$$\begin{aligned} \text{A.R.} &= |G(i\omega)| = \frac{K}{\sqrt{1 + \omega^2 \tau_1^2} \sqrt{1 + \omega^2 \tau_2^2}} \\ \phi &= \angle G(i\omega) = -\arctan(\omega \tau_1) - \arctan(\omega \tau_2) \end{aligned}$$

- When $\omega \ll 1/\tau_1$: A.R. $\rightarrow 1$, $\phi \rightarrow 0$
- When $\omega \gg 1/\tau_2$: A.R. $\propto \omega^{-2}$, $\phi \rightarrow -\pi$
- If $\tau_1 \gg \tau_2$ (time-scale separation), when $1/\tau_1 \ll \omega \ll 1/\tau_2$,
A.R. $\propto \omega^{-1}$, $\phi \approx -\pi/2$

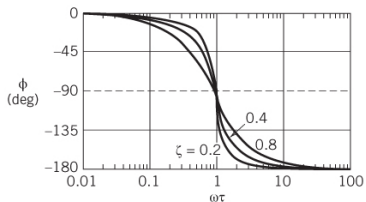
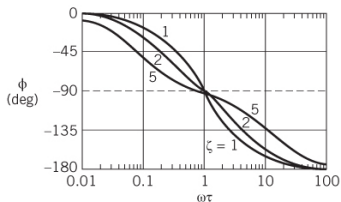
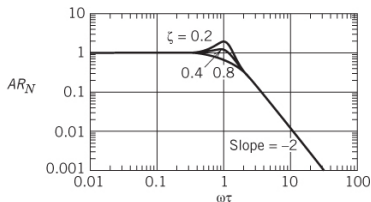
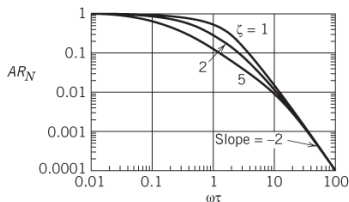
Example 7: Underdamped second-order systems

$$\begin{aligned} G(s) &= \frac{K}{\tau^2 s^2 + 2\tau\zeta s + 1} \quad (K > 0, 0 < \zeta < 1) \\ &= \frac{K}{(\tau s + \zeta + i\sqrt{1 - \zeta^2})(\tau s + \zeta - i\sqrt{1 - \zeta^2})} \end{aligned}$$

- Evaluate modulus and angle:

$$\begin{aligned} \text{A.R.} &= \frac{K}{|\zeta + i(\omega\tau + \sqrt{1 - \zeta^2})| \cdot |\zeta + i(\omega\tau - \sqrt{1 - \zeta^2})|} \\ &= \dots = \frac{K}{\sqrt{\omega^4\tau^4 + (4\zeta^2 - 2)\omega^2\tau^2 + 1}} \\ \phi &= -\arctan \frac{\omega\tau + \sqrt{1 - \zeta^2}}{\zeta} - \arctan \frac{\omega\tau - \sqrt{1 - \zeta^2}}{\zeta} \end{aligned}$$

- When $\zeta < \frac{1}{\sqrt{2}}$, at $\omega_r = \frac{\sqrt{1-2\zeta^2}}{\tau}$, A.R. = $\frac{K}{2\zeta\sqrt{1-\zeta^2}}$ reaches max.
- Such a phenomenon is called **resonance**



Higher-order systems: Calculating frequency response

- **Guideline:** Convert into a fractional T.F. and then **factorize** the numerator and denominator
- You should end up with a standard **gain/time-constant form**

Example 8: "Filtered PID controller"

$$G(s) = \frac{3}{0.5s + 1} \left(1 + \frac{1}{3s} + \frac{2}{3}s \right)$$

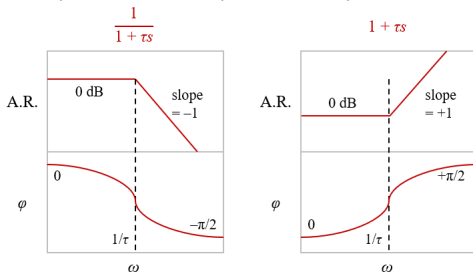
- ① Adding the fractions: $G(s) = \frac{3}{0.5s+1} \frac{3s+1+2s^2}{3s}$
- ② Factorize the terms: $G(s) = \frac{(2s+1)(s+1)}{s(0.5s+1)}$
- ③ Evaluate frequency response (term-wise)

$$\text{A.R.} = \frac{\sqrt{1+4\omega^2}\sqrt{1+\omega^2}}{\omega\sqrt{1+0.25\omega^2}}, \quad \phi = \arctan 2\omega + \arctan \omega - \frac{\pi}{2} - \arctan 0.5\omega$$

Higher-order systems: Sketching the Bode plot

$$G(s) = K \frac{(1 + \bar{\tau}_1 s) \dots (1 + \bar{\tau}_m s)}{(1 + \tau_1 s) \dots (1 + \tau_n s)} e^{-\theta s} \quad (K > 0)$$

- 1 If RHP zeros exist, first extract $(1 - \tau s)/(1 + \tau s)$ and plot for it (A.R.= 1, $\phi = -2 \arctan(\omega\tau)$)
- 2 Plot for every lead (in numerator) and lag (in denominator)



- 3 Plot for **gain** (A.R.= K , $\phi = 0$) and **delay** (A.R.= 1, $\phi = -\omega\theta$)
- 4 Add these curves together

Example 9: NMP, two lead, double lag

$$G(s) = 0.5 \frac{(1-s)(1+0.1s)}{(1+10s)^2}$$

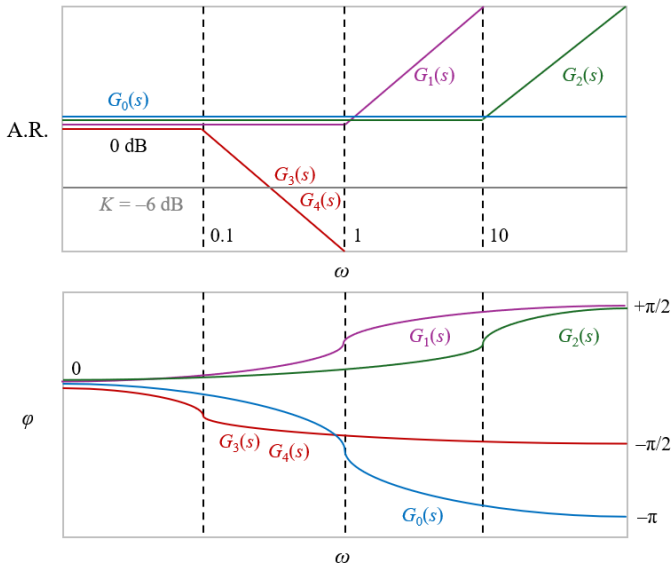
① Factorize!

$$G(s) = 0.5 \frac{1-s}{1+s} \frac{(1+s)(1+0.1s)}{(1+10s)(1+10s)} = KG_0(s) \frac{G_1(s)G_2(s)}{G_3(s)G_4(s)}$$

② Let's plot for each of them

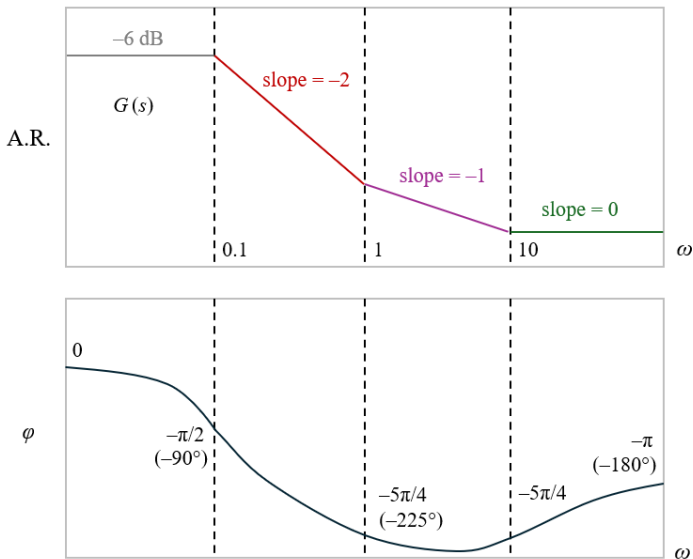
Higher-order systems: Sketching the Bode plot

- Let's plot for each of them



Higher-order systems: Sketching the Bode plot

- Let's add them together



Higher-order systems: Sketching the Bode plot

- The actual Bode plot

